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HARDWARE



Meta researchers are developing a gesture-controlled wristband that can interact with a computer



Meta researchers are [developing a wristband](#) that lets people control a computer using hand gestures. This includes moving a cursor, opening apps, and sending messages by writing in the air as if using a pencil.

Meta's wristband [employs a technique called surface electromyography \(sEMG\)](#), which detects electrical signals generated by muscle activity to interpret user movements, as explained in a research paper published in the journal *Nature*. These signals can sense a person's intended actions, even before they do them.

The goal of this wristband is to provide less invasive tools to interact with computers for people with motor disabilities.

The company is [collaborating with Carnegie Mellon](#) to test the wristband with people who have spinal cord injuries, enabling them to use computers even if they are unable to fully use their arms or hands. Douglas Weber, a professor in the Department of Mechanical Engineering and the Neuroscience Institute, noted that even people with complete hand paralysis still exhibit some muscle activity. As a result, the device is able to interpret their intended actions.

The wristband is a simpler alternative to other projects, such as Elon Musk's [Neuralink](#), which aims to implant brain chips in individuals with

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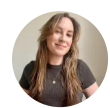
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severe paralysis. Other non-invasive approaches include headsets that utilize electroencephalogram (EEG) signals; however, these have limitations due to the weak signals they produce.

Meanwhile, Meta’s wristband could be used immediately without surgical intervention, and its use of sEMG signals means it operates at a higher frequency than EEG.

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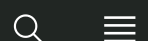
Lauren covers media, streaming, apps and platforms at TechCrunch.

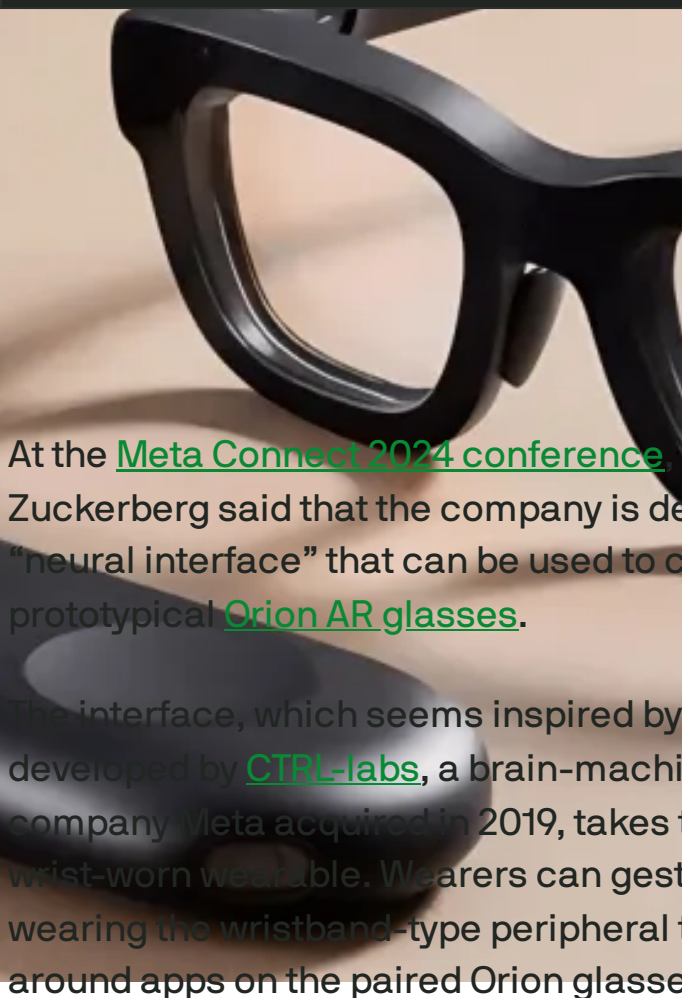
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Kyle Wiggers — 11:04 AM PDT · September 25, 2024

At the [Meta Connect 2024 conference](#), CEO Mark Zuckerberg said that the company is developing a “neural interface” that can be used to control its prototypical [Orion AR glasses](#).

The interface, which seems inspired by devices developed by [CTRL-labs](#), a brain-machine interface company Meta acquired in 2019, takes the form of a wrist-worn wearable. Wearers can gesture while wearing the wristband-type peripheral to navigate around apps on the paired Orion glasses.

IMAGE CREDITS: META

“[Orion is] the first device that is powered by our wrist-based neural interface,” Zuckerberg said. “[This is a] device that allows you to just send a signal from your brain to [Orion].”

The Verge’s Alex Heath [reports](#) that the wristband will soon go on sale and will work with Meta’s other AR hardware. Pricing has yet to be announced.

Meta’s Orion glasses — which remain very much a concept at this stage — are true AR. Orion utilizes tiny projectors built into the glasses’ temples to create a heads-up display.

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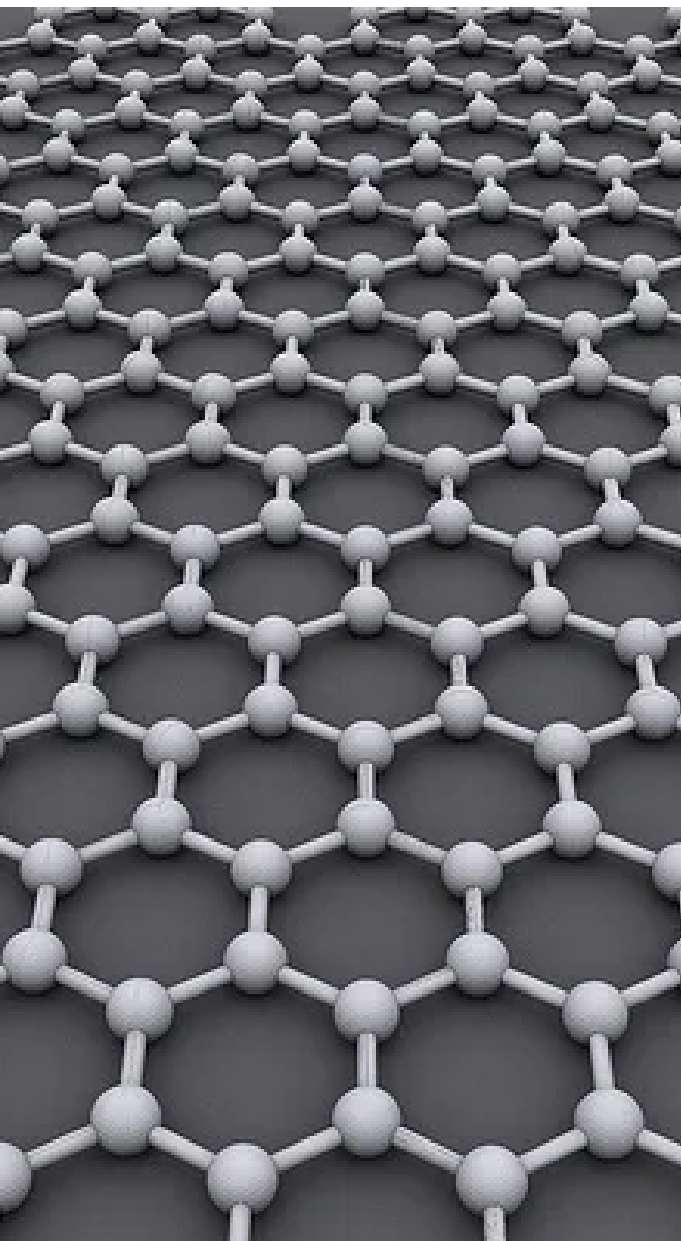
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Kyle Wiggers was TechCrunch's AI Editor until June 2025. His writing has appeared in VentureBeat and Digital...

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Graphene Shows Promise For Brain Implants

Natasha Lomas — 9:16 AM PST · February 7, 2016

[Graphene](#), the super thin carbon material that's been exciting scientists in the decade+ since single-atom thick graphene crystallites were successfully extracted from the bulk material, continues to give hints of a promising future blending electronics and biology.

In a new study, conducted by researchers at the [Cambridge Graphene Centre](#) and the University of Trieste in Italy, and published in the journal [ACS Nano](#), the suggestion is it could be used to make highly effective, flexible brain implants in future — biodevices that avoid the loss of signal problem associated with the scar tissue that can form around modern electrodes made from more rigid substances, such as silicon and tungsten.

Point is, human brains are made of soft tissue so it helps if your electrodes can flex too. Graphene is also considered to have excellent biocompatibility properties (although research into [potential toxicity](#) is not conclusive at this stage).

The implication of the Cambridge-Trieste research is that graphene-based electrodes could, in future, be safely be implanted in the brain — offering promise for the restoration of sensory functions for amputee or paralysed patients, for example, or to help individuals with motor disorders such as epilepsy or Parkinson's disease. So the future potential being glimpsed here is pretty exciting — albeit, theoretical and a long way out (plus, it should be stressed, the successful experiments were also conducted on rat brain cultures).

The researchers note that previously other groups have shown it is possible to use treated graphene to interact with neurons in the brain, however the problem with using treated graphene was the signal to noise ratio was very low. Working with untreated graphene retains the material's much

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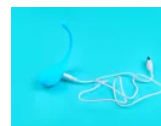
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lauded electrical conductivity — resulting in a significantly better electrode. And one that was seen to interface well with rat neurons.

“For the first time we interfaced graphene to neurons directly,” said Professor Laura Ballerini of the University of Trieste in Italy, in a statement. “We then tested the ability of neurons to generate electrical signals known to represent brain activities, and found that the neurons retained their neuronal signalling properties unaltered. This is the first functional study of neuronal synaptic activity using uncoated graphene based materials.”

The scientists couch the research as a “first step” towards using pristine graphene-based materials as an electrode for a neuro-interface. So again, graphene-based biodevices aren’t going to be coming to CES next year — perhaps in a couple of decades...



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They say their next steps will be to investigate how different forms of graphene are able to affect

neurons, and whether tuning the material properties might alter the biological response (in terms of synapses and neuronal excitability).

“Hopefully this will pave the way for better deep brain implants to both harness and control the brain, with higher sensitivity and fewer unwanted side effects,” added Ballerini.

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Natasha was a senior reporter for TechCrunch, from September 2012 to April 2025, based in Europe. She joine...

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